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Mapping the densities of malaria vectors within a single village

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Small scale spatial variation and temporal heterogeneity in mosquito densities can have important consequences for disease transmission, but the extreme variation which is observed in populations of malaria vectors makes it difficult to obtain good predictions of densities for short time periods over limited areas. We have applied Bayesian techniques derived for use in cancer epidemiology in order to map densities of *Anopheles gambiae s.l.* and *A. funestus* in a Tanzanian village where there is intense transmission of *Plasmodium falciparum* malaria. Estimates derived in this way should prove useful in vector population biology and in improving estimates of exposure–response relationships of the human host to malaria. The same methods can be applied in other fields of animal ecology.

Key words: Bayesian model; Gibbs sampling; Conditional autoregressive model; Spatial analysis; *Anopheles gambiae*; *Anopheles funestus*; Inoculation rate; Tanzania

Introduction

It difficult to avoid being bitten by anopheline mosquitoes in much of sub-saharan Africa, but densities of these malaria vectors within African villages vary widely both in space and time, sometimes in ways which are erratic and highly unpredictable. This introduces large errors into estimates of average biting densities when these are applied to small areas or for short periods.

In studies of malaria epidemiology in highly endemic areas, estimates of exposure to infectious bites have usually been made for entire villages (for instance in the classical study of Molineaux and Gramiccia, 1980) and/or for extended periods (for example, Burkot et al., 1987). This averaging out of the intra-village variation limits the scope of studies of exposure-response relationships. Heterogeneities in contact rates between mosquitoes and their hosts are important as causes of variation in the

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host population in morbidity, immunological status (for instance, Del Guidice et al., 1990) and parasitological responses. Such heterogeneities also have implications for models of malaria transmission. Mathematical models which allow for such heterogeneity generally lead to higher estimates of the basic reproductive rate for malaria than do those which ignore it (Dietz, 1988).

Understanding spatial and temporal variation in densities within villages would have immediate implications for inhabitants who wish to avoid anthropophilic mosquitoes. Does it matter where a house is built within the village? Or is it the mode of construction of the house which most affects the numbers of endophilic anophelines which will get inside? There is evidence that both factors have some importance in African villages (Gillies, 1988). Anopheline densities inside houses are affected by the presence of eaves (White, 1969; Snow, 1987; Lindsay and Snow, 1988), and the design of walls and windows (Schofield and White, 1984; Gillies, 1988). The presence of alternative hosts such as cattle can also be important, as in the Garki project (Molineaux and Gramiccia, 1980).

The location of houses can be important because of limited dispersal of mosquitoes from their breeding sites, but the processes which determine patterns of post-emergence dispersion in mosquito densities are difficult to measure and for present purposes essentially unobservable. Variation determined by such unobserved factors over and above sampling variation is termed *extra sampling variation* (Pocock et al., 1981). Data on measureable variables, such as architectural features, might be used to improve the precision of estimates of densities for different houses within a village, by treating them as covariates in a Poisson regression model (McCullagh and Nelder, 1989) for the numbers of mosquitoes. Such a model could be fitted by maximum likelihood using commercial software. However, standard methods cannot be used to model the structure of the extra sampling variation.

We assume that the mosquito densities are related to an underlying variable which cannot be measured directly, but whose value for any house, depends on that in neighbouring houses and hence shows some degree of spatial smoothness. These conditions are fulfilled by a conditional autoregressive (CAR) log linear model (Clayton and Kaldor, 1987; Besag et al., 1991; Breslow and Clayton, 1993). Such models have been successfully applied to the analysis of spatial variation in cancer incidence (Clayton and Kaldor, 1987; Bernardinelli and Montomoli, 1992; Breslow and Clayton, 1993).

Whilst maximum likelihood estimation of CAR models presents formidable problems, it is possible to estimate them within a hierarchical Bayesian framework, as proposed by Clayton and Kaldor (1987). Recent developments in the theory of Markov Chain Monte Carlo (MCMC) techniques (Geman and Geman, 1984; Gelfand and Smith, 1990), in computational algorithms (Wild and Gilks, 1993) and in computer software (Thomas et al., 1992) make it feasible to estimate the parameters of such models without analytical determination of likelihood functions.

We have now used this approach to estimate how densities of *Anopheles gambiae s.l.* and *A. funestus* varied geographically during the year 1990 in a single Tanzanian village, exploiting as much relevant data as possible on anopheline densities, house locations, and domestic environments. We describe how the results can be used in the analysis of the malariological impact of local spatial variation in entomological inoculation rate.

All these sources of variation need to be considered against a background of what

can be large and unpredictable night by night variations in mosquito population sizes. This also presents a problem for classical statistical methods. Fluctuations in the observed size of insect populations arises as a compounding of sampling variation and an underlying temporal process which is clearly related to meteorological factors, but in ways which defy formal time-series analysis and possibly involve chaotic processes (Logan and Allen, 1992). It is therefore unrealistic to expect to fit a statistical model for the underlying process leading to the temporal fluctuations. An alternative is to estimate a different parameter to allow for the population size on each night, avoiding the issue of how this variation arises. However, statistical models which require the unconstrained estimation of a different effect on mosquito numbers for each night do not allow for the sampling variation and therefore overestimate the inter-night variation in the underlying process. The maximum likelihood estimation of individual night effects is moreover impractical when the study extends over more than a few nights. In contrast, the hierarchical framework of the Bayesian model can be used to allow valid estimation of these night effects, by treating them as a sample from a distribution.

Methods

Study site

The study was carried out as part of the Kilombero Malaria Project (KMP) [1] in the village of Namawala located at the edge of the Kilombero valley in the Morogoro region of south eastern Tanzania (Tanner et al., 1991). Namawala village consists largely of numerous distinct groups of single-roomed houses, built with mud walls and thatched rooves, on the margins of the flood plain of the Kilombero river. There are few domestic animals: villagers are engaged in subsistence agriculture, cultivating maize, cassava and, in particular, rice grown in rain-fed paddies. Further details of the study area are provided by Smith et al. (1993) and Charlwood et al. (1995a,b).

House positions were recorded using a hand-held satellite navigational system (Trimble Navigational Europe Ltd, UK) as described by Snow et al (1993). This method gives positions to within ± 25m. The resulting map was validated against sketch maps of the village. The presence of windows in the houses, and of adjoining animal sheds were also recorded in surveys of the village.

Sampling of mosquitoes

The main anopheline species in Namawala are *Anopheles funestus*, and *A. gambiae s.l.* (including both *A. gambiae s.s.* and the morphologically indistinguishable *A. arabiensis*). These all contribute to the very high endemicity of *Plasmodium falciparum* malaria (Smith et al., 1993). Mosquito numbers were monitored by CDC light traps operated as described previously (Kilombero Malaria Project, 1992; Smith et al., 1993).

The present analyses refer only to mosquitoes collected during 1990. The main sampling scheme during this year involved 43 houses, each of which was targetted to be sampled once per fortnight (with the light traps being transferred between

houses during the fortnight). In practice, this sampling scheme was not followed rigidly, because of equipment failures and temporary migrations of householders. Where possible, nearby houses were sampled when target houses could not be sampled. Thus, a total of 63 houses were sampled.

In order to give an estimate of night by night changes in mosquito populations, one light trap was operated for a total of 275 nights during the year. This house is termed the sentinel house.

Statistical methods

Four different models were used to model factors affecting mosquito densities.

Fixed effects model The number of mosquitoes caught in house i on night j was treated as a Poisson variate with expectation $E_{i,j}$. Algebraically:

$$y_{i,j} \sim \text{Poisson}(E_{i,j})$$

Variation between houses was assumed to be independent, on the logarithm scale, from variation between nights (in other words, it is assumed that the spatial pattern does not show seasonality). This assumption was justified by the observation of near-proportional changes in mosquito numbers in a small number of houses sampled repeatedly (Charlwood et al., 1995a). The (natural) logarithm of $E_{i,j}$ was therefore analysed as a linear function:

$$\log E_{i,j} = \Psi_i + \Theta_j$$

Where Ψ_i is the effect associated with house i and Θ_j is the effect associated with night j. To make the parameters identifiable, Θ_1 was set to be 0. No further constraints on the Ψ_i or Θ_j values were imposed.

Random effects model In this model, the night specific parameters, Θ_j, were assumed to be drawn from a normal distribution, with mean 0 and precision γ. (Note that the precision is equal to the reciprocal of the variance).

$$\Theta_j \sim \text{Normal}(0,\gamma)$$

The house specific parameters, Ψ_i, were also assumed to be drawn from a normal distribution, this time with mean Ψ and precision τ

$$\Psi_i \sim \text{Normal}(\Psi,\tau)$$

The quantities γ,Ψ, and τ which describe the distributions of the vectors of parameters are known as hyperparameters. The choice of the normal family for these distributions was made after examining the distributions of parameters from the fixed effects model.

Mixed model The inter-house variation consisted of both extra sampling variation measured by the Ψ_i and effects associated with measured properties of the house. In the third model, parameters β and ω were introduced to estimate fixed effects corresponding to the two relevant measured properties of the house, the presence of an adjacent animal house and of window(s) respectively. The unmeasured variation between houses continued to be estimated by assuming the residual house

effects, Ψ_i, to be a sample from a normal distribution with mean Ψ and precision τ. The model was then:

$$\log E_{i,j} = \Psi_i + \beta b_i + \omega w_i + \Theta_j$$

where $b_i = 1$ if an animal shed was recorded adjoining house i; 0 otherwise; $w_i = 1$ if one or more windows were recorded in house i; 0 otherwise.

Conditional autoregressive (CAR) model A further model was fitted which treated the effects associated with measured properties of the house and with temporal variation the same way as described for the mixed model, but in addition a function ($\bar{\Psi}_{-i}$) was defined which measures the spatially structured variation. $\bar{\Psi}_{-i}$ was defined to be the mean of Ψ_i for all near neighbours of house i, not including house i itself. Proximity for this purpose was defined to correspond to a radius (r) of 150 m. In order to model short-range autocorrelation a small radius was chosen. Associations between houses which are further apart than r are then automatically allowed for. A lower limit to r is imposed because isolated houses (with no neighbours at distances $< r$) cannot be included in the model. The value $r = 150$ m was chosen because this was the smallest distance which ensured inclusion of almost all the houses sampled. (A total of 56 sampled houses, including the sentinel house, were thus included in the analysis. A further 7 sampled houses remained unconnected and were not included in the present analyses).

The effect of sampling within any one house (Ψ_i) was assumed to be normally distributed about $\bar{\Psi}_{-i}$. This variation arises partly because of imprecision in the estimation of $\bar{\Psi}_{-i}$ and partly because of unmeasured architectural and microenvironmental features of the house, including effects of the number and characteristics of its inhabitants. In order to allow appropriately for the imprecision in Ψ_{-i}, the precision of the normal distribution (τ_i) was assumed proportional to n_i, the number of near neighbours of house i.

$$\Psi_i \sim \text{Normal}(\bar{\Psi}_{-i}, \tau_i)$$

$$\tau_i = n_i \tau$$

It follows from these assumptions (Besag et al., 1991, Breslow and Clayton, 1993) that the joint distribution for the τ_i's has a simple form.

Model fitting The models were fitted by Gibbs Sampling (Geman and Geman, 1984; Gelfand and Smith, 1990) using the program BUGS (Thomas et al., 1992). Instead of calculating exact estimates, this computer intensive technique generates a stream of simulated values drawn from the distribution of each quantity of interest. The process is continued iteratively with results from the initial iterations before the estimates stabilise (the burn-in) being discarded. The quoted results were derived from a burn-in of 1500 iterations and estimation based on a further 500 samples. Diffuse but proper (mean 0, variance 10^5) normal prior distributions were used for the parameters μ, β, ω and diffuse (mean 1, variance 10^5) gamma priors for γ and τ. For each parameter 95% credible intervals were estimated. These intervals indicate ranges which include the true value of the parameter with a probability of 0.95.

Maps of mosquito densities A variable $\Psi_{x,y}$ was defined for all points (cartesian coordinates x,y) on the ground within 150 m of any sampled house. This was equal to the mean value of Ψ_i over all $n_{x,y}$ sampled houses within 150 m of point (x,y). $\Psi_{x,y}$ represents our best estimate of Ψ_k, the value of Ψ which would have been estimated for hypothetical house k, located at (x,y), had such a house been sampled. Values of $\Psi_{x,y}$ can therefore be used in order to estimate the mean log mosquito densities (per house) throughout the year for houses which were not sampled, providing such houses were located within 150 m of a sampled house. Contour plots of $\Psi_{x,y}$ were drawn using the software Mapinfo© (Mapping Information Systems, New York) in order to illustrate the spatial pattern in mosquito densities.

In addition to the sampled houses, a total of $m_{x,y}$ unsampled houses were recorded within 150 m of point (x,y). Assuming a two day gonotrophic cycle, half the adult female population attempts to feed on any one night. The light trap should catch all of those feeding. Therefore the predicted value for an average night, of the total number of mosquitoes per hectare in the circle of centre (x,y) (including those not feeding, and ignoring the effects of windows and animal sheds) was:

$$d_{x,y}=2 \exp(\Psi_{x,y})(m_{x,y}+n_{x,y})/2.25\pi$$

Contour plots of $d_{x,y}$ were drawn in order to determine the effect of housing density on the spatial pattern of mosquito densities.

Results

Data description

Fig. 1 shows the locations of both the 56 houses included in the analyses and other unsampled houses within Namawala (some of the houses were very close to each other and are therefore superimposed on the figure). Five of the sampled houses, including the sentinel house, were in the Kikulukutu area, some 3 km to the south west of the main part of the village (inset). Most of the houses in the main part of the village were on three low ridges running north-south. The valleys separating these ridges were inundated with water during the wet season.

The distribution of numbers of samples per house was highly skewed reflecting the difficulties encountered in implementing the original sampling plan. 8 houses were sampled only once, a total of 43 houses were sampled 15 or more times. Only one house was sampled more than 35 times. This was the sentinel house where a trap was operated on 275 nights during the year. Traps were operated on all but 80 nights during the year, but on 196 nights only one trap was operated. On 46 nights more than 10 different light traps were operated. The numbers of mosquitoes caught in the sentinel house varied widely (Fig. 2; see also Smith et al., 1993) indicating substantial fluctuations in population sizes. It follows that the database is extremely unbalanced and that crude estimates of mosquito densities obtained by taking marginal averages over nights or over houses would be very unreliable.

Overall densities of *A. funestus* were very similar to those of *A. gambiae* (Table 1). Table 1 also compares the mean logarithms and standard deviation of the logarithms of numbers of mosquitoes caught in houses with animal sheds and those without. An equivalent comparison of samples from houses with and without windows is also shown. No major effect of either windows or animal sheds was evident in either

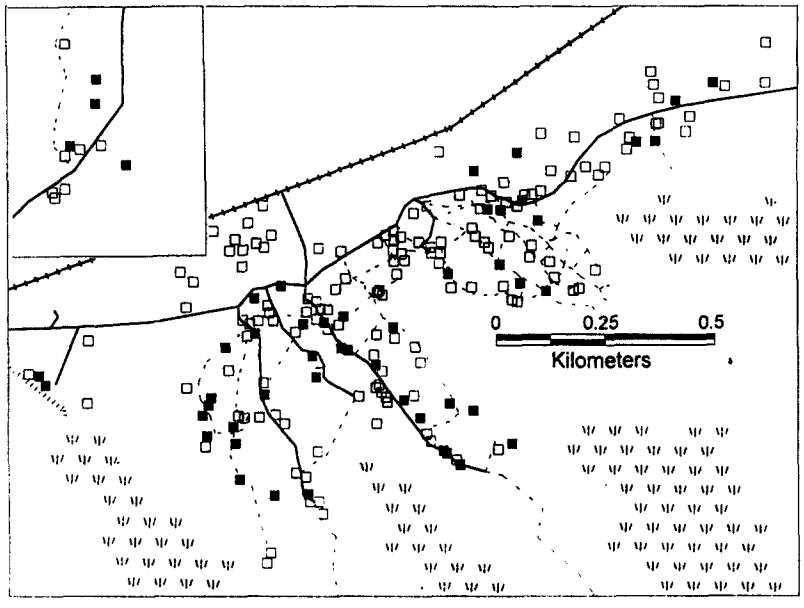


Fig. 1. Map of Namawala village. ●: sampled houses; □: other houses; ++++: railway; — — : road; – – –: path; \¦/: swamp. The inset refers to an area (Kikulukutu) approximately 3 km south west of the main part of the village.

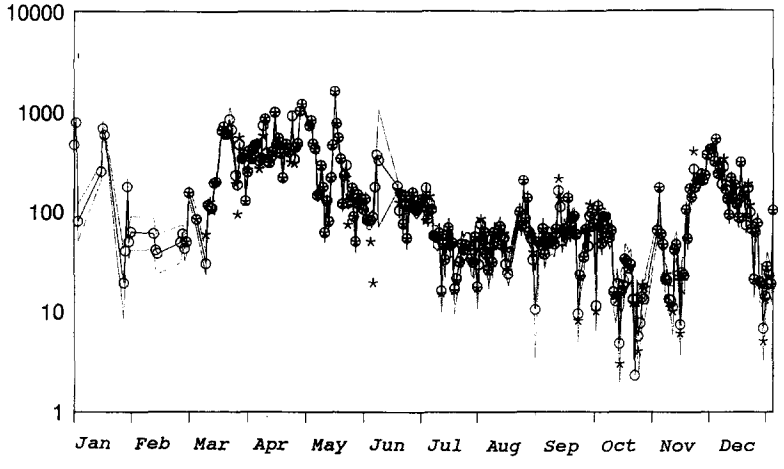


Fig. 2. Observed and predicted numbers of mosquitoes caught in the sentinel house by night during 1990. ×: Numbers of *A. gambiae* sampled in the sentinel house by night; 0: Numbers of *A. gambiae* in the sentinel house predicted from the CAR model. Dotted lines join the upper and lower 95% credible limits.

species. However, because of the lack of balance in the data, it is necessary to adjust for the confounding effects of time trends and position in the village before we can know whether these are real effects of these microenvironmental features, since there are likely to be biases caused by the uneven allocation of sampling effort. It is clearly important to allow for this lack of balance in the dataset before making inferences about these factors.

TABLE 1

Relationship of numbers of mosquitoes caught with domestic microenvironment

	No. of houses	No. of samples	*A. gambiae s.l.* $\log_e(y_{ij}+1)$		*A. funestus* $\log_e(y_{ij}+1)$	
			Mean	S.D.	Mean	S.D.
No animal shed recorded	49	490	1.05	1.40	1.05	1.27
Animal shed recorded	13	183	1.31	1.56	1.00	1.43
No window recorded	13	80	1.09	1.42	0.73	1.12
Window recorded	49	593	1.12	1.45	1.07	1.32
All houses	62	673	1.12	1.45	1.04	1.31

Note that data for the sentinel house are omitted from this table, but that the 7 isolated houses excluded from the main analysis are included here.

Statistical models

The four different models which were estimated for the *A. gambiae* data differed in the way in which the variation between houses was handled. In a previous paper (Smith et al., 1993) we suggested that a fixed effects log-linear model overestimates the variation between houses. In Fig. 3a the estimates of the house effects obtained from such a model are compared with those from the random effect model. After adjustment for the average level, there was generally good agreement between the parameters estimated from these two models for each house, suggesting that the assumption of an underlying normal distribution was reasonable. Houses with very low or zero *A. gambiae* recorded, were an exception to this. The random effects model estimated very low but non-zero mosquito densities for most of these houses. The exceptions illustrate situations where the random effects model gives better estimates. For instance, house $i=53$ was sampled only once, and no *A. gambiae s.l.* were caught in it. The fixed effects model therefore estimated $\Psi_{53}=-\infty$ for this house. Eleven other houses were sampled the same night as this house but *A. gambiae s.l.* were caught in only one of them. The random effects model estimated $\Psi_{53}=-0.024$ with 95% credible interval -2.72 to 2.06 which correctly indicates that we have little evidence that this house has densities very different from the average for the village.

The house effects estimated by the random effects (Ψ_i) and the mixed models ($\Psi_i+\beta b_i+\omega w_i$) were very similar. The CAR model estimated a reduced level of variation between houses in the underlying densities (Ψ_i) implying that more of the variance was assigned to sampling variation. This is illustrated in Fig. 3b. The estimates for houses with relatively large numbers of samples are similar in the random effects and CAR models. However, estimates for poorly sampled houses were pulled towards those of their neighbours (and hence away from the random effects estimates) because of the spatial smoothing, a phenomenon known as shrinkage. In general this meant that the CAR estimates were closer to the overall mean. However, one house, with a single observation of zero mosquitoes had an exceptionally low fitted density in the CAR model ($\Psi_i+\beta b_i+\omega w_i=-2.26$). This estimate was influenced by the location of this house in a low mosquito density area of the village, and its possession of an animal shed.

In each of the three models where a random effect was used to model the inter-

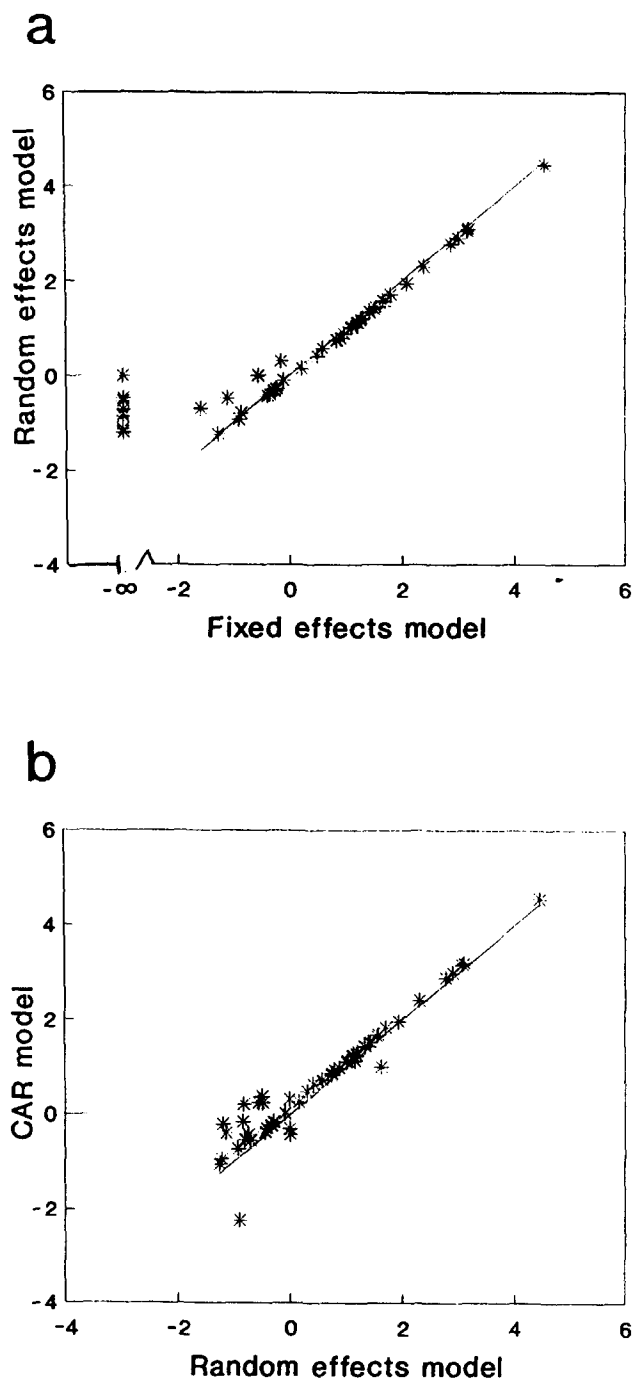


Fig. 3. House effects estimated from different models for *A. gambiae s.l.* (a) Comparison of Ψ_i estimated from fixed effects and random effects models (b) Comparison of Ψ_i estimated from random effects model with $\Psi_i + \beta b_i + \omega w_i$ estimated from CAR model. Each point corresponds to one sampled house. The solid line is the line of identity on which all points would fall if both estimates were the same.

night variation, almost identical estimates of the night effects were obtained. The estimates of the mean and precision of the distribution of night effects are given in Table 2. Fig. 2 shows how the *A. gambiae s.l.* density in the sentinel house predicted from the CAR model compares with the observed numbers caught in the light trap. For 193 out of the 285 nights when sampling was carried out, only the trap in the sentinel house was operated, therefore it is not surprising that a good fit was observed for these nights in this house. However, on only a small number of the remaining nights did the observed sample size fall outside the credible interval. Such discrepancies are to be expected occasionally because of sampling variation. Therefore the modelling of inter-night variation was satisfactory.

The modelling of the effects of windows and of animal houses implied that the data descriptions in Table 1 were affected by confounding of house construction with geographical location. Table 2 shows that the presence of a window was associated with a 2% increase in estimated *A. gambiae s.l.* densities when adjusted for the positional effect using the CAR model, but the credible interval was from an 8% decrease to 11% increase. In contrast, *A. funestus* densities were reduced by 62% (cred. int. 57%–66%) in houses with windows. Animal sheds were associated with decreases in the estimates for both species (*A. gambiae s.l.*, 53% (cred. int. 23%–71%), *A. funestus*, 62% (cred. int. 57%–66%)). We did not investigate the possibility that there might be an interaction between the two effects.

The precision of the night effects was lower for *A. funestus* than for *A. gambiae* (Table 2). The precision of 0.70 for *A. gambiae s.l.* corresponds to a variance of 1.43 in the night effects (compare Table 3). If we define normal fluctuations to correspond to variation within limits of ± 2 standard deviations of the mean, this implies that night by night variation of the order of 4.8 on a natural log scale, corresponding to changes in mosquito densities by a factor of 116. The corresponding calculation for *A. funestus* indicates population variation by a factor of 1300.

The precision parameter for the spatial effects (τ) is related to the inter house variance in log densities only via the numbers of neighbours (n_i) and hence does not easily give an estimate of the range of variation between houses in densities. The empirical variance (Table 3) of 1.67 for *A. gambiae s.l.* corresponds to variation

TABLE 2

Parameter estimates from CAR models

Parameters	*A. gambiae s.l.*			*A. funestus*		
	Mean	95% cred.	interval	Mean	95% cred.	interval
ψ Mean of house effects	0.86	0.67	1.05	1.16	0.96	1.34
β Effect of animal shed	−0.75	−1.23	−0.26	−0.41	−0.66	−0.19
ω Effect of window	0.02	−0.08	0.10	−0.96	−1.08	−0.84
Hyperparameters						
γ Precision of distribution of night effects	0.70	0.59	0.83	0.31	0.25	0.36
τ Precision of distribution of positional effects	0.25	0.15	0.39	0.16	0.10	0.26

Note that the data for the sentinel house are omitted from this table, but that the isolated houses excluded from the main analysis are included here.

TABLE 3

Variance components of the linear predictor for *A. gambiae s.l.*

Source of variation	Model	Variance estimate	% Of variance	95% Credible interval
Nights	Random[a]	1.42	43.4	35.2–50.1
Nights	Mixed[a]	1.42	44.1	36.1–50.0
Nights	CAR[a]	1.42	46.2	37.2–52.2
All inter-house	Random[b]	1.87	56.6	49.9–64.8
All inter-house	Mixed[c]	1.82	55.9	50.0–63.9
All inter-house	CAR[c]	1.67	53.8	47.8–62.9
Unmeasured inter-house	Mixed[b]	1.83	56.4	49.8–64.8
Unmeasured inter-house	CAR[b]	1.63	52.6	47.0–60.5
Spatial inter-house	CAR[d]	0.89	28.8	23.2–34.2

[a] Empirical variance of θ_j.
[b] Empirical variance of Ψ_i.
[c] Empirical variance of $\Psi_i + \beta b_i + \omega w_i$.
[d] Empirical variance of $\bar{\Psi}_{-i}$.

in mosquito densities between houses (± 2 standard deviations on the log scale) by a factor of 176.

Components of variance of the fitted log transformed mosquito densities

The observed distributions of the parameter estimates from the models were used to estimate the contribution of different sources to overall variation in (log-transformed) expected numbers of *A. gambiae s.l.* (Table 3). Inter-night variation accounts for 46.2% of the variance as estimated from the CAR model, and a little less as estimated from the other mixed model and the random effects model. This is because the spatial smoothing results in a rather reduced estimate of inter house variation in the CAR model. Only a small percentage of the inter-house variation can be attributed to the measured factors. In fact, in the mixed model, the variance of Ψ_i was less than that of $\Psi_i + \beta b_i + \omega w_i$. In the CAR model using the difference between these quantities as an estimate of the contribution of the measured factors to the variance, a total of 1.2% of the variance could be attributed to the shed or windows.

Spatial inter house variation accounted for about half the inter-house variance.

Mapping of mosquito densities

Fig. 4a is a contour plot of the function $\Psi_{x,y}$ for *A. gambiae s.l.* illustrating the pattern of *A. gambiae s.l.* densities by house, over the village allowing for the inter-night variation. The region for which $\Psi_{x,y}$ and $d_{x,y}$ are defined includes the sites of most of the unsampled houses in the village. For each of these houses, it is therefore possible to use the CAR model to estimate mosquito densities. The areas which are not shaded on the maps are those for which $\Psi_{x,y}$ cannot be determined because the nearest sampled house was more than 150m distant. This does not mean that there are no mosquitoes there. Areas of high mosquito density as indicated in this map,

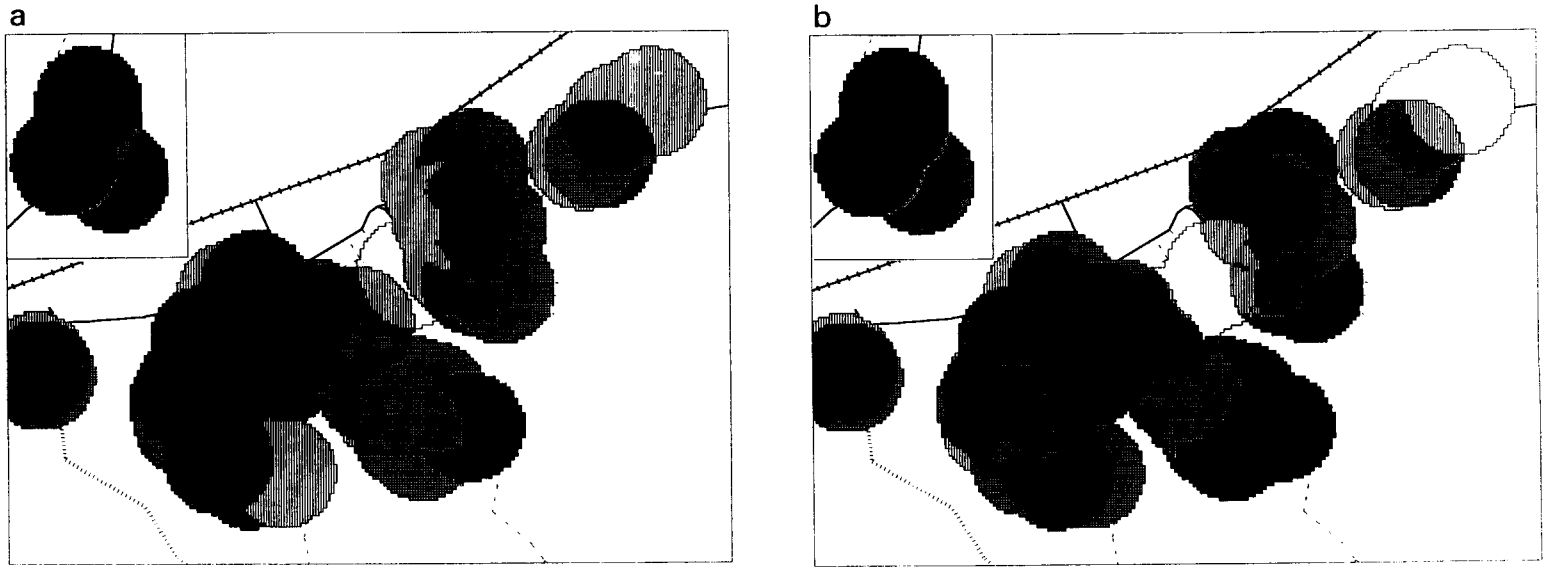


Fig. 4. Contour plot of mean log mosquito densities per house ($\Psi x,y$) in Namawala village during 1990 (a) *A. gambiae s.l.*. (b) *A. funestus*. The inset again refers to the Kikulukutu area.

were predominantly those nearest the low lying areas in which rice was grown, including the fringes of the streams running through the main part of the village. The overall pattern for *A. funestus* was similar (Fig. 4b), with an even stronger tendency for the mosquitoes to be concentrated at the southern edge of the village.

Fig. 5a is a corresponding plot of the $d_{x,y}$ for *A. gambiae s.l.*, the estimated density per hectare, allowing for the different housing density in different parts of the village. Despite the higher mosquito density in the peripheral houses, the higher housing density in the centre of the village more than compensated, so that most of the mosquitoes are in fact feeding on the denser populated ridges. *A. funestus* (Fig. 5b) showed less tendency to concentrate in the centre of the village and densities continued to be highest on the southern edges of the ridges where they border the rice-fields.

Discussion

In field studies in African villages, practical considerations make it difficult to sustain a balanced random sampling scheme for mosquitoes. Simple average mosquito densities can therefore give misleading estimates of the pattern of variation. Regression methods are needed to allow appropriately for non-random sampling of mosquitoes in the village.

Because of sampling error, the usual (fixed effects) regression approaches exaggerate the amount of local variation in mosquito densities when used to estimate mosquito densities for short periods or for small geographical areas. This provides the justification for using random effects models. The lack of balance also means that there is considerable variability between houses in the credibility which can be attached to any of the estimates of mosquito numbers. This can be illustrated by the results for the house with only a single observation for which very different Ψ_i estimates were obtained by the fixed and random effects model. The estimate from the random effects model with its wide credible limits, appropriately indicates both the considerable uncertainty attached to the estimate for this house, and also conforms to the common sense inference that our best guess is that this house has a similar density to the average for the village. Random effects models therefore give better estimates than fixed effect models.

An important feature of the Gibbs Sampling algorithm applied to any of the models is that it provides estimates of the posterior distribution for all these densities. These distributions can be used in order to ensure that appropriate allowance is made for the variation in credibility of the estimates. For instance, when averaging exposures for groups of houses, or in the determination of entomological inoculation rates (EIR) or exposure response relationships, instead of using point estimates of mosquito densities, a Gibbs sampler can be used to sample mosquito numbers from the posterior distributions estimated from the model.

The conditional autoregressive model is a way of further improving estimates of mosquito densities and of obtaining appropriate estimates with credible intervals for each house in the village and for each night on which sampling was carried out. Whilst the inclusion of the autoregressive terms in the model did not have much effect on most of the estimates for sampled houses, it did make it possible to attach estimates (albeit with lower credibility) to houses which were not sampled, but which

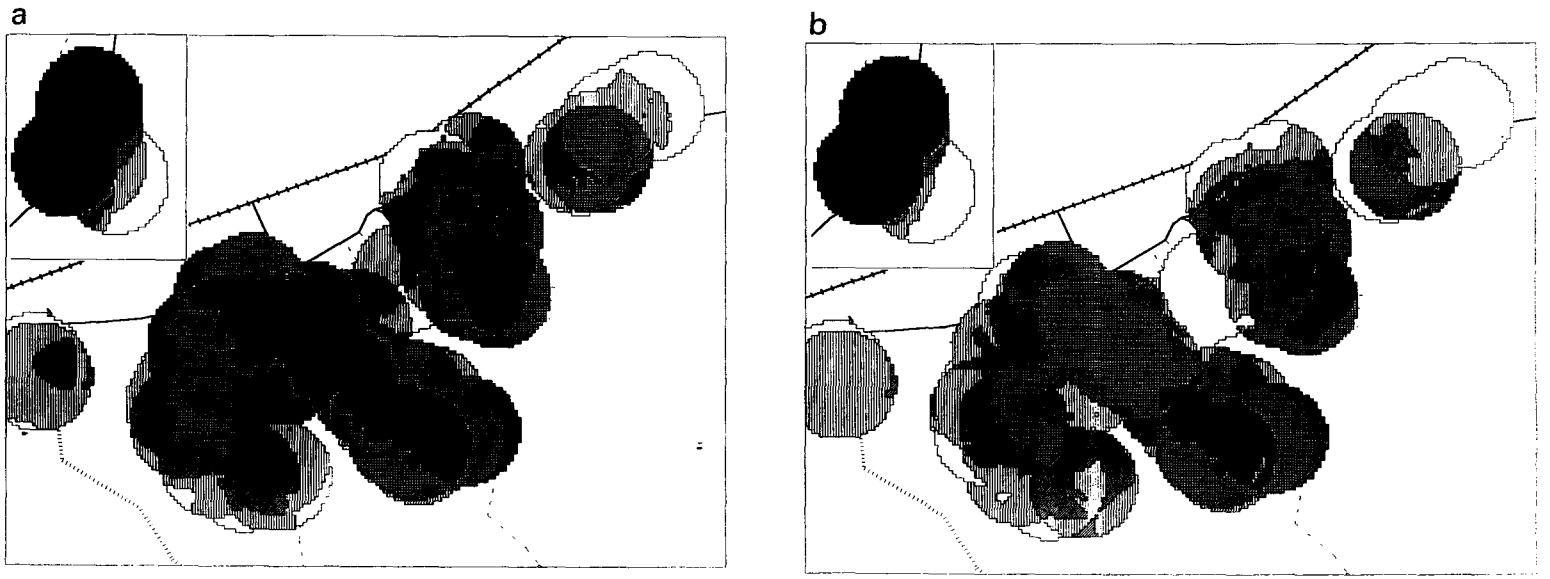


Fig. 5. Contour plot of estimated mosquito densities per hectare ($d_{x,y}$) in Namawala village on an average day during 1990 (a) *A. gambiae s.l.*. (b) *A. funestus*.

were located within the 150 m radius of sampled houses. Since a substantial proportion of the variation between houses in inoculation rates was found to be accounted for by the spatial process, it is not unreasonable to make use of these positional data in estimates of mosquito densities. Unfortunately we still cannot provide clear advice to the villager who wants to avoid mosquitoes. Moving his/her house within the village will not have a very profound effect on mosquito numbers.

Constructing the house without windows was associated with little effect on *A. gambiae s.l.* densities in Namawala. In theory windows allow the entry of more mosquitoes, and in this context, the finding (Table 2) that *A. funestus* densities were reduced in houses with windows was surprising, especially since the opposite effect was suggested by Table 1. This may reflect association of the windows with other features which were not studied. The houses without windows were concentrated on the ridges with low *A. funestus* densities which can account for the lower crude densities in windowless houses.

Animal sheds appear to have reduced the number of mosquitoes somewhat, which is what might be expected, since animals residing next to the house might result in some mosquitoes (particularly the less anthropophilic *A. arabiensis*) being diverted away to alternative hosts. However we did not have data on the occupancy of these sheds, so these data alone do not provide strong evidence for an effect of nearby alternative hosts on biting rates.

The modes of construction of most of the study houses were very similar but many of the newer houses in Kilombero are being constructed from brick rather than the traditional unbaked mud. We were not able to analyse the effect of such radically different architectures on mosquito numbers because there was insufficient variation in construction technique among the sampled houses. It is not surprising therefore that most of the non-spatial inter-house variation was accounted for by unmeasured factors, the most obvious of which might be variation in the number and age distribution of both human and animal inhabitants. It is well known that hungry mosquitoes are attracted to houses in densities that are related to the numbers of human occupants (Gillies, 1955).

The maps illustrate a spatial pattern of average mosquito densities per house which corresponds well with the impression gained by the entomological staff in the field. There were few mosquitoes in the higher ground in the northern part of the village. Areas of high mosquito density tended to follow the valleys running north south through the village, where breeding sites are most persistent. Densities were highest in the low-lying area of Kikulukutu. The pattern of higher anopheline densities in the peripheral zones is similar to that found in urban communities elsewhere in Africa (Trape and Zoulani, 1987; Trape et al., 1992). This probably reflects a tendency of the mosquitoes to try to feed in the first house which they encounter on leaving the breeding sites.

It was less obvious in the field how the mosquitoes were distributed per unit area. Gillies (1955) used spray catching of indoor resting mosquitoes to estimate densities of *A. gambiae s.l.* in a village in northern Tanzania. These were found to range seasonally from 5/hectare to 120/hectare. *A. funestus* densities ranged from 1.25 to 195/hectare. The densities at the centre of Namawala were comparable to these, but our results imply a greater degree of seasonality in Kilombero, with again more variation in *A. funestus* densities than *A. gambiae s.l.*. However, the variation between different areas in the village was comparable in magnitude to this seasonal variation.

Despite the higher densities in the peripheral houses, the majority of the *A. gambiae s.l.* were feeding in the core of the village, with Kikulukutu representing an exception to this general pattern. This is consistent with the opportunistic breeding habits of this species group, and illustrates the effectiveness with which it locates human dwellings. *A. funestus* showed a stronger tendency to concentrate in the houses on the edges of the rice fields (Fig. 4b), and this pattern to some extent still held when the densities are converted to mosquitoes per hectare. This probably reflects this species' requirement for more permanent water in its breeding sites (Gillies and De Meillon, 1968).

Gibbs sampling is computationally intensive (some of the models required several hours processing time on a 33 MHz 486 PC), and requires special software. However, since both fast computers and this software are likely to become increasingly available, this will not represent a major limitation to the application of these techniques in the future.

The approach which we have used in this paper will be applicable in many further analyses of mosquito densities. The same statistical methods are applicable to data from resting or landing catches of mosquitoes. Extension of the model to allow seasonal variation in Ψ_i will enable us to rigorously test whether the mosquito populations in different houses vary proportionately throughout the year. From contour maps of $d_{x,y}$ for an area as a whole it might also be possible to delimit areas occupied by independent populations, particularly in the dry season. Since the mosquitoes are endophilic, and the positions of all the houses are known, summation of fitted values from such a model can be used to estimate absolute population sizes which could be compared with effective population sizes obtained from genetic analysis (Taylor et al., 1993).

Estimates of mosquito densities can be combined with malaria sporozoite prevalences in order to obtain entomological inoculation rates for malaria (EIR). Because of the contribution of sampling variation to inter-house variability in mosquito densities it has generally been considered unrealistic to obtain EIR estimates at the individual household level. Studies of exposure response relationships in malaria have therefore assumed mosquito densities for a limited number of sites within a village to be representative of the whole village and have used EIR estimates based on these houses to be applicable to the whole village. The levels of inter-house variation indicated by our analysis imply that the choice of sentinel households can make a large difference to the estimates which would be obtained from such a procedure. The obvious choice of sampling indicator houses in the centre of the village would lead to underestimation of the overall exposure to mosquitoes in Namawala. By sampling many houses and weighting the household specific estimates of mosquito densities from the CAR model by the human population in each house, we are now able to compute an overall average EIR which allows for the spatial heterogeneity. Moreover, we can compute household specific EIR estimates which will make it possible to determine the contribution of exposure differences to inter-household heterogeneity in immunoepidemiological parameters.

The general approach is not only applicable in study of malaria vectors. Bayesian mapping approaches might well be applicable in many other analyses of geographical variation in epidemiological measurements where conventional statistical methods cannot adequately allow for the spatial autocorrelation structure of the data.

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